

- (24) Michels, A.; Sengers, J. V.; Van der Gulik, P. S. *Physica (Amsterdam)* **1962**, *28*, 1216-1237.
- (25) Sheldon, S. J.; Knight, L. V.; Thorne, J. M. *Appl. Opt.* **1982**, *21*, 1663-1669.
- (26) Carter, C. A.; Harris, J. M. *Appl. Opt.* **1984**, *23*, 476-481.
- (27) Hirschfelder, J. O.; Curtiss, C. F.; Bird, R. B. "Molecular Theory of Gases and Liquids"; Wiley: New York, 1954; Chapters 4 and 5.
- (28) Jackson, W. B.; Amer, N. M.; Boccara, A. C.; Fournier, D. *Appl. Opt.* **1981**, *20*, 1333-1344.
- (29) Fournier, D.; Boccara, A. C.; Badoz, J. *Appl. Opt.* **1982**, *21*, 74-46.
- (30) Carter, C. A.; Harris, J. M. *Appl. Spectrosc.* **1983**, *37*, 166-172.
- (31) Isaacs, Neil S. "Liquid Phase High Pressure Chemistry"; Wiley: New York, 1981; Chapter 6.
- (32) Carter, C. A.; Harris, J. M. *Anal. Chem.* **1984**, *56*, 922-925.
- (33) Buffet, C. E.; Morris, M. D. *Appl. Spectrosc.* **1983**, *37*, 455-458.
- (34) Moldover, M. R.; Sengers, J. V.; Gammon, R. W.; Hocken, R. J. *Rev. Mod. Phys.* **1979**, *51*, 79-99.
- (35) Harris, J. M.; Leach, R. A.; Hardcastle, F., presented in part at the Pittsburgh Conference and Exposition, Atlantic City, NJ, March 1984; Paper 595.

RECEIVED for review January 17, 1984. Accepted March 23, 1984. This material is based upon work supported by the National Science Foundation under Grant CHE82-06898.

Statistical Uncertainties of End Points at Intersecting Straight Lines

Lowell M. Schwartz* and Robert I. Gelb

Department of Chemistry, University of Massachusetts, Boston, Massachusetts 02125

Procedures are described for calculating the statistical uncertainty of a titration end point defined by the intersection of straight line segments. This type of titration curve often occurs in analytical techniques such as conductometry, spectrophotometry, and amperometry. Equations are derived for the standard error estimate and the confidence interval for the end point. Also discussed are statistical uncertainties of end point difference assays, i.e., assays calculated from the difference of two end points, both of which are defined by intersecting straight lines. An illustrative example shows calculational details, including the effect of data points which deviate from the straight segments and the effect of increasing noise level.

Segmented linear titration curves are often encountered in routine chemical analysis. For example, this type of curve is observed in Gran plots and in titrations monitored by conductometry, spectrophotometry, and amperometry employing dropping mercury or rotating platinum indicator electrodes. In these techniques, deviations from linearity are often observed directly at the end point. Other techniques, such as the so-called "dead-stop" or "biamperometric" titration, yield curves which feature linear segments contiguous to both sides of the end point but deviations occur elsewhere. We are not aware of a previously published discussion of the statistical uncertainty of end points obtained from linear segmented titration curves and so in this paper will develop procedures for assigning these uncertainties. These procedures are based on the statistical fluctuation in discrete titration data. Thus titration data must be available in digital form or, if the original titration curve is recorded in continuous (analog) form, discrete points whose values include a random sampling of the noise fluctuation must be selected.

Titration curves of this nature consist of two or more branches each of which consists partly of a linear portion. We will denote the coordinates of two adjacent branches as y_A vs. x_A and y_B vs. x_B . The end point defined by these two branches is the intersection of the two straight segments. In many conductometric, spectrophotometric, and amperometric titrations this intersection lies beyond the linear ranges of both. The end point of interest is the abscissa value of this inter-

section, which we will denote by X . Once the parameters of the two lines are found, it is a simple matter to calculate X algebraically. In order to facilitate the upcoming statistical analysis, we will write the two linear segments as

$$y_A = \bar{y}_A + b_A(x_A - \bar{x}_A) \quad (1a)$$

$$y_B = \bar{y}_B + b_B(x_B - \bar{x}_B) \quad (1b)$$

where $(\bar{x}_A, \bar{y}_A)$ and $(\bar{x}_B, \bar{y}_B)$ are the centroids of the A and B linear segments, respectively. The intersection occurs when $y_A = y_B$ and when $x_A = x_B = X$. Application of these conditions to eq 1a and 1b yields

$$X = -(\bar{y}_B - \bar{y}_A + b_A\bar{x}_A - b_B\bar{x}_B)/(b_B - b_A) \equiv -\Delta a/\Delta b \quad (2)$$

for the end point. The notations Δa and Δb represent the differences in y intercepts and slopes of the two lines, respectively.

One problem we pose in this paper is to estimate statistical uncertainties for X . Our approach to this problem depends on the statistical natures of discrete observations of the y and x . If the relative statistical uncertainties of the x data are negligible compared to the y data, the method of least squares is appropriate and is the most common choice. It has been shown (1, 2) that the method of least squares is also the correct method to use even when the x data have substantial errors provided that these data are experimental settings as contrasted with experimental measurements. In order to illustrate this difference, consider a titration done by using an ordinary manual buret. If the incremental volumes x are recorded by reading the positions of the meniscus on the graduated scale, these volumes are subject to randomly varying reading errors. This means that if the titration could be replicated in such a way that the same volume increments are transferred, the recorded volumes x would differ from replicate to replicate because of the reading errors. Volume data obtained in this way are random variables. On the other hand, suppose the experimenter decides to transfer titrant in increments of 1 mL. After setting the initial meniscus position to zero or to some other integer milliliter level, he proceeds to transfer titrant by 1 mL at a time, stopping to record the response datum and the integer volume setting. He does not examine the level of meniscus after each transfer but rather is satisfied that 1 mL has been transferred as long as the meniscus is reasonably near an integer mark. There is no doubt that the

actual volumes transferred in this manner differ from the recorded volumes, but these errors are not reflected in the x data as recorded. If this titration is replicated, the lists of recorded volumes for all replications are identical; all are the same list of integer milliliter volumes. These recorded x values are *not* random variables. The distinction between random and nonrandom variables will be important in the following discussion. In this paper we will treat only those titrations carried out in a manner such that the method of least squares is properly applied. This means either that the x values are nonrandom because they are settings or that random x values have errors which are negligibly small compared to the random errors in the response variables.

The statistical uncertainty of X depends on the uncertainties of the two linear segments which are reflected in the variances of the four parameters $\bar{y}_A$, $\bar{y}_B$, b_A , and b_B . The parameter variances in turn depend upon the variances of the ordinate values which we denote as vary y or as σ^2 . We will entertain the possibility that var y is not uniform along the titration curve but varies from point to point with the magnitude of the response y . In other words, we consider the ordinate variance, in general, to be a function of the position of the point along the titration curve and this function $\sigma^2(y, x)$ can be expressed in terms of either y or x . This function may be completely specified or it may involve an unknown constant multiplying factor V . This means that we may know that

$$\sigma^2(y, x) = V\phi(y, x) \quad (3)$$

where $\phi(y, x)$ is a completely specified function of x or y .

For each line segment the method of least squares yields parameters $\bar{y}$ and b according to the following equations (3). Here we denote the individual observations along the titration curve by the subscript i . There are n_A such observations in the linear segment A and n_B observations in segment B. In the interest of conciseness we have deleted the subscripts A and B which distinguish the two line segments, but these subscripts are understood to apply to all quantities.

$$b = S_{xy}/S_{xx} \quad (4a)$$

where

$$S_{xy} = \sum_i w_i (x_i - \bar{x})(y_i - \bar{y}) \quad (4b)$$

$$S_{xx} = \sum_i w_i (x_i - \bar{x})^2 \quad (4c)$$

$$\bar{x} = (\sum_i w_i x_i) / \sum_i w_i \quad (4d)$$

$$\bar{y} = (\sum_i w_i y_i) / \sum_i w_i \quad (4e)$$

and

$$w_i = 1/\sigma^2(y_i, x_i) \quad (4f)$$

The sums are taken from $i = 1$ to n_A or n_B as appropriate. If the variance function is uniform, $\sigma^2(y, x)$ is a constant and these expressions reduce to the familiar equations for the ordinary unweighted least-squares method.

How we find the variances of the least-squares parameters $\bar{y}$ and b depends on our knowledge of the ordinate variance function. If $\sigma^2(y, x)$ is completely specified, the parameter variances are (ref 3, section 11.12)

$$\text{var } b = 1/S_{xx} \quad (5a)$$

$$\text{var } \bar{y} = 1/\sum_i w_i \quad (6a)$$

However, if $\sigma^2(y, x)$ involves an unknown multiplier as expressed by eq 3, this multiplier must be calculated from the residual scatter of the data points about the two least-squares linear segments. The procedure is as follows: (ref 3, section

11.13): (1) calculate b and $\bar{y}$ for both lines by using eq 4a and 4e but instead of using eq 4f for the weighting factors, use

$$w_i = 1/\phi(y_i, x_i) \quad (4g)$$

(2) Calculate weighted sums of squared residuals, S_A and S_B , for each of the line segments according to

$$S = \sum_i \{[\bar{y} + b(x_i - \bar{x}) - y_i]^2 / \phi(y_i, x_i)\} \quad (7)$$

(3) Pool the variance estimates from both line segments into a single V by using the following formula which weights each contribution by the appropriate number of degrees of freedom

$$V = \sum_j [(n_j - 2)S_j] / \text{df} \quad (8a)$$

where

$$\text{df} = \sum_j (n_j - 2) \quad (8b)$$

is the number of degrees of freedom inherent in V . The summations are with respect to all lines, here $j = A$ and $j = B$. (4) Having calculated V in this manner, the line parameter variances are

$$\text{var } b = V/S_{xx} \quad (5b)$$

$$\text{var } \bar{y} = V/\sum_i w_i \quad (6b)$$

where w_i are taken according to eq 4g both in eq 6b and in S_{xx} of eq 5b.

UNCERTAINTY ESTIMATES FOR THE END POINT

If the standard deviations of the ordinate data are a small fraction of their magnitudes, a standard error estimate for X can be calculated by applying propagation-of-variance procedures (4) to eq 2. The end point X depends on four least-squares line parameters, $\bar{y}_A$, $\bar{y}_B$, b_A , and b_B , which are random variables, but no pair of them is statistically correlated. Segment A parameters depend only on measurements made along segment A and these are statistically independent of the measurements along B. Also the parameters $\bar{y}$ and b for any given line can be shown to be uncorrelated (ref 3, section 11.3). $\bar{x}_A$ and $\bar{x}_B$ are not random variables. However, both Δa and Δb are correlated random variables because each involves b_A and b_B . First-order propagation-of-variance retains only first derivatives in the Taylor expansions and this procedure leads to

$$\text{var } \Delta a = \text{var } \bar{y}_B + \text{var } \bar{y}_A + \bar{x}_B^2 \text{var } b_B + \bar{x}_A^2 \text{var } b_A$$

$$\text{var } \Delta b = \text{var } b_B + \text{var } b_A$$

$$\text{cov}(\Delta a, \Delta b) = -\bar{x}_B \text{var } b_B - \bar{x}_A \text{var } b_A$$

$$\Delta b^2 \text{var } X = \text{var } \Delta a + X^2 \text{var } \Delta b + 2X \text{cov}(\Delta a, \Delta b)$$

and eventually to

$$\text{var } X = [\text{var } \bar{y}_B + \text{var } \bar{y}_A + \sum_j (\bar{x}_j^2 - 2\bar{x}_j X + X^2) \text{var } b_j] / \Delta b^2 \quad (9a)$$

where $\sum_j$ again means a sum of two terms, one with subscript $j = A$ and one with $j = B$. The standard error estimate of X is

$$s_X = (\text{var } X)^{1/2} \quad (9b)$$

The standard error s_X predicts the standard deviation of X that would be observed if identical replicate titrations were done. An alternative way of expressing the statistical uncertainty of X is by means of confidence limits. This statistic, however, depends on the distribution function of the random variable X . If the ordinates y_i are assumed to have normal

(Gaussian) distributions, the least-squares parameters as well as Δa and Δb of eq 2 are also normally distributed (ref 3, section 12.4). However, X , even if regarded as the ratio of two normally distributed variables, is not normally distributed and, indeed, becomes more and more skewed as the variance levels increase. For sufficiently small variance though, X is approximately normally distributed and under these conditions confidence limits may be calculated from s_X which is also accurate only when variances are small. These confidence limits are $X \pm t_\alpha s_X$, where t_α is the Student's t statistic at the $(1 - \alpha)$ confidence level and for the number of degrees of freedom inherent in s_X . This number is given by eq 8b if the multiplier V was calculated from the least-square line residuals or is taken as infinite if the variance function $\sigma^2(y, x)$ is known a priori.

We now turn our attention to the calculation of confidence limits when the variance of the responses are not necessarily small. The assumption of normally distributed y values is retained, however. This development closely parallels a similar derivation of confidence limits for assays derived from calibration curves (5, 6). Imagine that we carry out many replicate titration experiments and rather than compute an end point for each titration, we calculate least-squares lines for all the A segments and for all the B segments. The mean of all the A slopes is $\langle b_A \rangle$ and the mean of all the B slopes is $\langle b_B \rangle$. Similarly $\langle \bar{y}_A \rangle$ and $\langle \bar{y}_B \rangle$ are the mean y centroids of the two sets of lines. The "best" end point may be computed from this collection of data as that X which satisfies the equation

$$\langle \bar{y}_A \rangle + \langle b_A \rangle (X - \bar{x}_A) = \langle \bar{y}_B \rangle + \langle b_B \rangle (X - \bar{x}_B)$$

This X is the abscissa of intersection of the two "average" line segments. Now consider any A, B pair of individual line segments written as a difference ϵ as follows

$$\epsilon = [\bar{y}_B + b_B(X - \bar{x}_B)] - [\bar{y}_A + b_A(X - \bar{x}_A)] = \Delta a + X \Delta b \quad (10)$$

For any such pair of lines the difference ϵ is not, in general, zero because the "best" end point cannot be the end point for each pair of lines in the collection. However, the mean $\langle \epsilon \rangle$ of all these ϵ values is zero and ϵ is normally distributed because it is formed as a linear combination of normally distributed variables. If $\text{var } \epsilon$ is the variance of this ϵ , the ratio $\epsilon/(\text{var } \epsilon)^{1/2}$ is distributed as Student's t , or $\epsilon^2/\text{var } \epsilon$ is distributed as t^2 . Thus if t_α is Student's t for the $(1 - \alpha)$ confidence level, upper and lower confidence limits for X can be calculated as the roots of the quadratic equation

$$\epsilon^2 = t_\alpha^2 \text{var } \epsilon \quad (11)$$

where ϵ is given by eq 10 and $\text{var } \epsilon$ by

$$\text{var } \epsilon = \text{var } \Delta a + X^2 \text{var } \Delta b + 2X \text{cov}(\Delta a, \Delta b) \quad (12)$$

Again the number of degrees of freedom inherent in $\text{var } \epsilon$ is either infinite or given by eq 8b.

As was discussed in the case of calibration curve assays (5, 6), there will be pathological situations for which two meaningful roots of the quadratic equation cannot be found. This will tend to occur as the slopes of the two line segments approach each other and as the variances of the responses increase. These pathological situations become evident when eq 11 is plotted in the following way. After substituting ϵ of eq 10 into the square root of eq 11, we can arrange the result in the form

$$\bar{y}_A + b_A(X - \bar{x}_A) = \bar{y}_B + b_B(X - \bar{x}_B) \pm t_\alpha (\text{var } \epsilon)^{1/2} \quad (13)$$

The left-hand side of eq 13 plotted vs. X is the segment A least-squares straight line. The right-hand side of eq 13 also plotted vs. X has two branches which lie above and below the segment B least-squares line. The intersections of the segment A line with these two branches represent solutions to equation

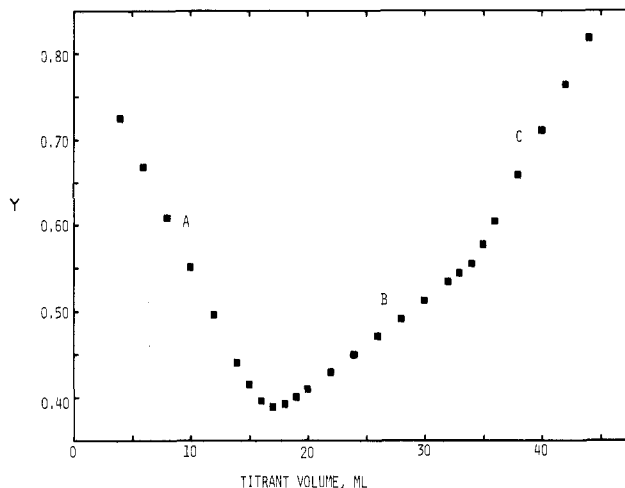


Figure 1. Illustrative example: conductometric curve obtained by titrating a mixed acid solution with KOH. Plotted data point coordinates are given in Table I.

11 or 13 and the abscissa values of these intersections are the confidence limits of the end point X . In the pathological situations two intersections do not occur, and when this happens, meaningful confidence limits do not exist, at least not at the $(1 - \alpha)$ confidence level.

ILLUSTRATIVE EXAMPLE

In order to show how these calculations are done, we will analyze the data recorded from a conductometric titration of a 100-mL mixed aqueous solution of perchloric and acetic acids with 0.1000 F standardized potassium hydroxide solution. We made conductance measurements with a Leeds and Northrup Model 4959 conductance bridge equipped with a dip-type conductance cell and operating at 1000 Hz. From a 50-mL buret, we added equal 1-mL volume increments of standardized base but did not actually take care to add exactly 1.00 mL at a time. Rather we added approximately this amount. Incremental volume data taken in this manner should be treated as settings as described above. We recorded a total of 45 data points which spanned all three branches of the titration curve; the first (branch A)—neutralization of perchloric acid, the second (branch B)—neutralization of acetic acid, and the third (branch C)—increasing concentration of excess KOH. In part I of four parts to this illustrative example, we seek an assay of the perchloric acid only. A plot of raw conductance data ($1/R$) vs. volume (x) is not, in general, expected to be linear because of the dilution effect of the titrant and so, as is the usual practice, we plot the product $y \equiv (1/R)(100 + x)$ vs. x and this is shown in Figure 1. By examining this plot we have selected 13 points which appear to fall on the linear segments of branches A and B, $n_A = 6$ and $n_B = 7$ points, respectively. The raw data and ordinate values are given in the top section of Table I. Data points near the perchloric acid end point on the y vs. x plot obviously deviate from the straight line segments and these points are marked with a superscript d in Table I. These particular points are not used in part I but are discussed in part II of this example. Because the conductance measurements span such a narrow range of values, we will assume that the conductance data all have the same constant, but unknown, variance which we denote by V . The y_i data, which are the product of conductance and $(100 + x_i)$, do not all have the same variance, however. The y_i variances are, in fact, $V(100 + x_i)^2$. Therefore, the function ϕ of eq 3 is $(100 + x_i)^2$ and the weighting factors of eq 4g are $(100 + x_i)^{-2}$.

The lower part of Table I shows intermediate results so that an interested reader can follow the computational procedures in detail. The end point X is found to be at 16.37 mL and

Table I. Illustrative Example: Conductometric Titration of a 100-mL Mixed Acid Solution with 0.1000 F KOH

Raw Data and Ordinate Values								
branch A			branch B			branch C ^e		
1/R ^a	x _A ^b	y _A ^c	1/R	x _B	y _B	1/R	x _C	y _C
6.975	4	0.7254	3.330 ^d	18	0.3929	4.280	35	0.5778
6.305	6	0.6683	3.370 ^d	19	0.4010	4.445	36	0.6045
5.638	8	0.6089	3.420	20	0.4104	4.772	38	0.6585
5.020	10	0.5522	3.522	22	0.4297	5.080	40	0.7112
4.432	12	0.4964	3.633	24	0.4505	5.380	42	0.7640
3.865	14	0.4406	3.742	26	0.4715	5.680	44	0.8179
3.610 ^d	15	0.4152	3.840	28	0.4915			
3.415 ^d	16	0.3961	3.946	30	0.5130			
3.328 ^d	17	0.3894	4.052	32	0.5349			
			4.097 ^e	33	0.5449			
			4.145 ^e	34	0.5554			

Illustrative Example Part I: Computational Details

$$n_A = 6$$

$$\Sigma w_A = 6.065 \times 10^{-4}$$

$$\bar{x}_A = 8.786$$

$$\bar{y}_A = 0.5881$$

$$n_B = 7$$

$$\Sigma w_B = 4.423 \times 10^{-4}$$

$$\bar{x}_B = 25.746$$

$$\bar{y}_B = 0.4690$$

$$X = 16.366 \text{ mL}$$

$$b_A = -0.02854$$

$$S_A = 6.003 \times 10^{-10}$$

$$b_B = 0.01037$$

$$S_B = 2.019 \times 10^{-10}$$

$$V = 3.790 \times 10^{-10}$$

$$\text{var } \bar{y}_A = 7.482 \times 10^{-7}$$

$$\text{var } b_A = 6.424 \times 10^{-8}$$

$$\text{var } \bar{y}_B = 8.569 \times 10^{-7}$$

$$\text{var } b_B = 5.365 \times 10^{-8}$$

$$s_X = 0.081 \text{ mL}$$

$$\text{var } \Delta a = 4.216 \times 10^{-5}; \text{cov } (\Delta a, \Delta b) = -1.946 \times 10^{-6}; \text{var } \Delta b = 1.179 \times 10^{-6}$$

$$\text{quadratic equation 11: } 1.513 \times 10^{-2} X^2 - 0.04953 X + 0.4052 = 0$$

$$95\% \text{ confidence limits: } 16.550 \text{ and } 16.183 \text{ mL; } 95\% \text{ CI} = 0.367 \text{ mL}$$

^a Conductance, $\text{k}\Omega^{-1}$. ^b Titrant volume, mL. ^c $y = (1/R)(100 + x) \Omega^{-1} \text{ mL}$. ^d These data are used only in part II calculations. See Table II. ^e These data are used only in part IV calculations. See Table IV.

so the assay result is 1.637 mmol of HClO_4 . The variance V is computed from the scatter of the 13 points from their appropriate straight line segments. The value $V = 3.790 \times 10^{-10}$ implies that the effective standard deviation of the conductance measurements is $(3.790 \times 10^{-10})^{1/2} = 1.95 \times 10^{-5}$ and that the coefficient of variation based on an average conductance value of 0.0045 is 0.43%. This effective standard deviation reflects both the inherent scatter of the conductance measurements and the scatter of these values from the titration curve due to the discrepancies between the actual volumes transferred and the recorded volume settings. The standard error estimate of the end point is 0.081 mL, which when compared to the end point value, yields a coefficient of variation of 0.49%. We then calculate 95% confidence limits and so use $t_\alpha = 2.262$, which corresponds to the 9 degrees of freedom in the variance V . The confidence limits are 16.550 and 16.183 mL and their difference, 0.367 mL, is the 95% confidence interval (CI). Figure 2 shows the graphical representation of this calculation in the form of eq 13. We may compute an effective standard error estimate from the confidence interval as follows. If the variances were small enough that the random variable X were normally distributed, the confidence interval would be $2t_\alpha s_X$. Whether or not this approximation is valid, we define the effective standard error estimate $(s_X)_{\text{eff}}$ such that

$$\text{CI} = 2t_\alpha (s_X)_{\text{eff}} \quad (14)$$

We can compare $(s_X)_{\text{eff}}$ with s_X to see the effect of nonnormality in X . In this example, the effective standard error estimate is $0.367/(2 \times 2.262) = 0.0811 \text{ mL}$, which is virtually identical with s_X . The noise level in this experiment is so small that the small-variance approximation is perfectly adequate. We shall explore higher noise levels in part III of this example.

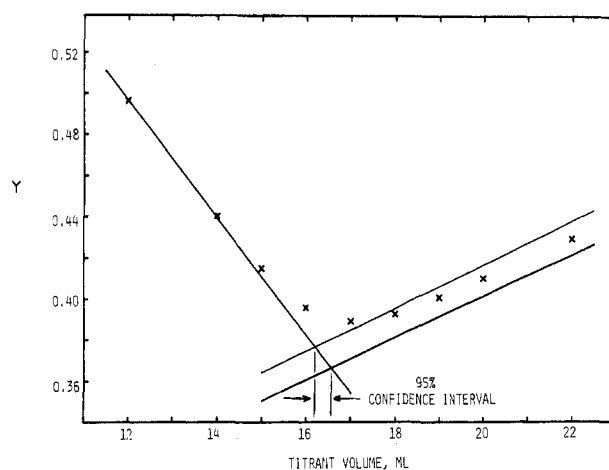


Figure 2. Illustrative example part I: graphical representation of eq 13 calculation of 95% confidence limits for the perchloric acid end point.

Part II deals with the problem of deciding which data points are to be included in the straight line segments of the two branches of the titration curve. Some data points near the end point may appear to deviate from straight line extrapolations fitted to points further away, but it is not always clear whether or not to omit these problematical points from the analysis. There may be competing effect on the uncertainty estimates for the end point: If the questionable points are omitted, fewer points are used and these span a narrower range of x values. This tends to increase the parameter variances. Also t_α increases when V is calculated with fewer degrees of freedom. On the other hand, if the questionable points are true deviants, their inclusion increases all variances if V is

Table II. Illustrative Example Part II: The Effect of Data Points Near the End Point

inclusive data points		n_A	n_B	CI, mL
x_{Ai} , mL	x_{Bi} , mL			
4-15	20-32	7	7	0.491
4-14	20-32	6	7	0.367
4-12	20-32	5	7	0.356
4-10	20-32	4	7	0.423
4-12	19-32	5	8	0.391
4-12	20-32	5	7	0.356
4-12	22-32	5	6	0.353 (optimal)
4-12	24-32	5	5	0.483

Table III. Illustrative Example Part III: The Effect of Increasing Scatter of the Conductance Data

conductance ^a		95% confidence limits, mL	s_X , mL	CI/ ($2t_{\alpha}s_X$)
variance	cv, %			
1V	0.43	16.55, 16.18	0.081	1.000
5V	0.97	16.78, 15.95	0.182	1.001
10V	1.4	16.95, 15.78	0.257	1.002
50V	3.1	17.68, 15.05	0.575	1.010
100V	4.3	18.24, 14.48	0.813	1.021
500V	9.7	20.93, 11.73	1.819	1.118
1000V	14	23.78, 8.77	2.573	1.289
2000V	19	34.11, -2.45	3.638	2.221

^a $V = 3.790 \times 10^{-10}$ as calculated in Part I. cv = average coefficient of variation = (variance)^{1/2}/0.0045.

calculated from the data. If the variance function of y is completely known a priori, however, the effect of including deviants is mainly to introduce systematic error, which shifts the slopes of the line and biases the X determination. In this example, where we do calculate V from the data, we will explore the effect of problematical points on the end point uncertainty. The optimal set of data points to use in the analysis is that set which yields the minimum value in some uncertainty statistic which we choose here to be the 95% confidence interval. We start with the set used in part I, for which CI = 0.367 mL, and after adding the doubtful point at 15 mL, find that these 14 points yields CI = 0.491 mL. Thus we conclude that the point at 15 mL is, indeed, off the A line. Next we eliminate the two points at 15 and 14 mL and find CI = 0.356 mL, which is better than our original calculation in part I. When we eliminate the next point at 12 mL, CI

increases to 0.423 mL. We conclude that the five points, 4-12 mL, are the best set for line A. Fixing these five points on line A, we then explore the inclusion and exclusion of points on the B line and find the results shown in Table II. The optimal configuration is five points (4-12 mL) on A and six points (22-32 mL) on B. These data yield the minimum CI of 0.353 mL and an end point of 16.351 mL. While it is true that in this example the optimal end point value and its uncertainty are statistically equivalent to those found by selecting the points by eye as done in part I, it is useful to see how the optimization can be done. It is a simple matter to automate the optimization process by digital computer code.

In part III we show how the standard error estimate of equation 9b deviates from the confidence interval as the magnitude of the conductance variance increases. We do this by increasing V from the value found in part I up to a factor of 2000 times as shown in Table III. All the other data used in part I remain unchanged. The second column in Table III lists the average coefficient of variation corresponding to each variance in the first column. The final column shows the ratio of the effective standard error (eq 14) to the small-variance standard error estimate (eq 9b). We see that the small-variance standard error always underestimates the effective standard error. The deviation is a modest 2% when the coefficient of variation of the data is near 4% but the discrepancy increases rapidly at higher noise levels. At coefficients of variation greater than about 20% the quadratic equation 11 fails to yield two real roots and so confidence limits cannot be found at the 95% confidence level.

STATISTICAL UNCERTAINTY OF END POINT DIFFERENCES

We now turn to the problem of assaying the acetic acid in the mixed acid solution. The end point near 34 mL, which occurs between branches B and C of the titration curve shown in Figure 1, corresponds to the complete neutralization of both acids by the titrant KOH. Thus if X_1 is the volume at which the straight lines A and B intersect and X_2 is the volume at which the B and C lines intersect, the difference $X_2 - X_1$, which we will denote by ΔX , when multiplied by the formality of the titrant, is the molar quantity of acetic acid in the mixture. The assay itself is found easily by applying eq 1 and 2 separately to the A, B lines and the B, C lines. However, the statistical uncertainty of ΔX is not a simple combination of uncertainties for X_1 and X_2 . An attempt to derive equations for exact confidence limits for ΔX by using the reasoning which leads to eq 11 and 12 fails here because the quantity analogous to ϵ of eq 10 involves products of random variables.

Table IV. Illustrative Example Part IV: Optimizing the Calculation of ΔX

inclusive data points			n_A	n_B	n_C	$2t_{\alpha}s_{\Delta X}$
x_{Ai} , mL	x_{Bi} , mL	x_{Ci} , mL				
4-12	22-32	38-44	5	6	4	0.862
4-12	22-32	36-44	5	6	5	0.549
4-12	22-32	35-44	5	6	6	0.424
4-12	22-32	34-44	5	6	7	0.684
4-12	22-33	35-44	5	7	6	0.373
4-12	22-34	35-44	5	8	6	0.356 (optimal)
4-14	22-34	35-44	6	8	6	0.482

Some Details of the Optimal Calculation

$$X_1 = 16.358 \text{ mL}$$

$$V = 0.9453 \times 10^{-10}$$

$$\text{var } X_1 = 26.44 \times 10^{-4}$$

$$\text{cov}(\Delta a_1, \Delta a_2) = 2.9836 \times 10^{-2}$$

$$\text{cov}(\Delta a_1, \Delta b_2) = -3.5281 \times 10^{-2}$$

$$X_2 = 34.245 \text{ mL}$$

$$df = 13$$

$$\Delta X = 17.887 \text{ mL}$$

$$t_{0.05} = 2.160$$

$$\text{var } X_2 = 60.95 \times 10^{-4}$$

$$\text{cov}(\Delta a_2, \Delta b_1) = -1.6853 \times 10^{-2}$$

$$\text{cov}(\Delta b_1, \Delta b_2) = 2.0349 \times 10^{-2}$$

$$\Sigma_4 \text{cov} = -19.50 \times 10^{-4}$$

$$\text{var } \Delta X = 67.89 \times 10^{-4}$$

small-variance 95% confidence limits for ΔX : 18.065, 17.709 mL

Therefore, this quantity is not normally distributed and so exact confidence limits cannot be found in terms of Student's t distribution. On the other hand, first-order propagation of variance applied to ΔX is feasible and this procedure leads to a standard error expression for ΔX analogous to eq 9b for a single end point. In the following equations, subscripts 1 and 2 refer to the end points between branches A, B and B, C, respectively. The end point difference

$$\Delta X = -\Delta a_2/\Delta b_2 + \Delta a_1/\Delta b_1 \quad (15)$$

is a random variable depending explicitly on four random variables, the Δa 's and Δb 's as shown. Because all four of these variables depend on the data points which lie along branch B, nonzero covariances exist between all four. Therefore, the variance of ΔX is the sum of variances of X_1 and X_2 as expressed separately by eq 9a plus the four covariances

$$\text{cov}(\Delta a_1, \Delta a_2) = 2(\text{var } \bar{y}_B + \bar{x}_B^2 \text{var } b_B)/\Delta b_1 \Delta b_2 \quad (16a)$$

$$\text{cov}(\Delta a_1, \Delta b_2) = -2\bar{x}_B X_2 \text{var } b_B/\Delta b_1 \Delta b_2 \quad (16b)$$

$$\text{cov}(\Delta a_2, \Delta b_1) = -2\bar{x}_B X_1 \text{var } b_B/\Delta b_1 \Delta b_2 \quad (16c)$$

$$\text{cov}(\Delta b_1, \Delta b_2) = 2X_1 X_2 \text{var } b_B/\Delta b_1 \Delta b_2 \quad (16d)$$

The standard error estimate is

$$s_{\Delta X} = (\text{var } X_1 + \text{var } X_2 + \sum_4 \text{cov})^{1/2} \quad (17)$$

where $\sum_4 \text{cov}$ is the sum of eq 16a-d.

We now apply these equations to the analysis of the complete titration curve of the illustrative example seeking to find an optimal set of data points for the acetic acid assay. Again following the logic of part II, we include and exclude data points falling near both end points in a search for the minimum uncertainty in ΔX . Because we cannot calculate exact confidence limits in this case, we use the small-variance confidence interval $2t_{\alpha} s_{\Delta X}$ instead. The results of this search

appear in Table IV. We start with the optimal data set for X_1 , which is five points $x_{Ai} = 4-12$ mL and six points $x_{Bi} = 22-32$ mL, and find that six points $x_{Ci} = 35-44$ mL, yields a provisional minimum confidence interval of 0.424 mL. Then with this set of six points fixed on branch C, we add one point at a time to branches A or B as shown in the table. The final optimal set yields a minimum CI of 0.356 mL. Some of the details of this optimal calculation are also shown in Table IV. A ΔX value of 17.887 mL leads to an acetic acid assay of 1.789 mmol which has 95% confidence limits of 1.806 and 1.771 mmol.

It is interesting to note that the optimal 95% confidence interval for ΔX (0.356 mL) appears to be almost the same as that for X_1 alone (0.353 mL). These two intervals are based on different variance V estimates, however. A better comparison can be made when we put both intervals on the same basis by replacing $V = 1.670 \times 10^{-10}$ ($df = 7$, $t_{\alpha} = 2.365$) with $V = 0.9453 \times 10^{-10}$ ($df = 13$, $t_{\alpha} = 2.160$) in the X_1 statistics. When this is done, the 95% confidence interval for X_1 reduces to 0.243 mL. The variance of ΔX is less than the sum of the variances of X_1 and X_2 in this example because the net effect of the covariance terms is subtractive. This fact cannot be expected to be true in general, however.

LITERATURE CITED

- (1) Berkson, J. J. *Am. Stat. Assoc.* **1950**, *45*, 164-180.
- (2) Acton, F. S. "Analysis of Straight-Line Data"; Dover Publications: New York, 1966; pp 50-53.
- (3) Brownlee, K. A. "Statistical Theory and Methodology in Science and Engineering", 1st ed.; Wiley: New York, 1960.
- (4) Kendall, M.; Stuart, A. "The Advanced Theory of Statistics", 4th ed.; Macmillan: New York, 1977; Volume 1, Section 10.6.
- (5) Schwartz, L. M. *Anal. Chem.* **1977**, *49*, 2062-2068.
- (6) Schwartz, L. M. *Anal. Chem.* **1979**, *51*, 723-727.

RECEIVED for review February 3, 1984. Accepted March 23, 1984.

Analysis of Skin Lipids for Halogenated Hydrocarbons

Mary S. Wolff

Environmental Sciences Laboratory, Mount Sinai School of Medicine of The City University of New York,
1 Gustave L. Levy Place, New York, New York 10029

Utilization of skin lipid analysis has been investigated as a noninvasive alternative to blood or adipose for estimation of human body burden of persistent halogenated hydrocarbons. For p,p' -DDE (2,2-bis(4-chlorophenyl)-1,1-dichloroethane), which was determined in 110 paired skin lipid and blood serum samples and in 29 concomitant adipose samples, the skin lipid analysis provided an acceptable alternative to adipose or serum. The method was less satisfactory for other residues, which were observed at lower concentrations than p,p' -DDE, including hexachlorobenzene, *trans*-nonachlor, and polychlorinated biphenyls (PCBs). However, the results suggested that with sufficiently long sample collection time, the method would be useful. The amount of skin lipids collected was greater with longer sampling time, as was the amount of halogenated hydrocarbon residue.

Human exposure to chemicals may be derived from the workplace or from other environmental sources. Estimation

of chemicals in the body, based on pharmacological considerations, provides an important biological marker in clinical evaluation or in epidemiologic studies which establish population norms of human exposure. Chemicals or their metabolites are usually determined as concentrations in blood, urine, maternal milk, or adipose tissue. Adipose tissue has been of particular importance since lipophilic pesticide residues, which are poorly metabolized, are largely sequestered in lipid, with partition via blood to other tissues. Partition of such chemicals to milk occurs during lactation, facilitated by the fat content of milk and by efficient blood flow to mammary tissue (1).

Utilization of noninvasive techniques to estimate chemicals in the body has been of interest to us as a means of evaluating human exposure. We have investigated skin lipids as an accessible source of determining chemical body burden.

The skin is rich in lipids and is a major storage site of halogenated hydrocarbons in animals (2). Excretion of skin oil (or sebum) has been reported as a quantitative phenomenon, in the context of dermatological pathogenesis (3, 4). Matthews and co-workers have reported the analysis of halogenated hydrocarbons in lipids from human hair (5). We